

Research

Invasion of the Hawaiian Islands by a parasite infecting imperiled stream fishes

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Points of origin and pathways of spread are often poorly understood for introduced parasites that drive disease emergence in imperiled native species. Co-introduction of parasites with non-native hosts is of particular concern in remote areas like the Hawaiian Islands, where the introduced nematode *Camallanus cotti* has become the most prevalent parasite of at-risk native stream fishes. In this study, we evaluated the prevailing hypothesis that *C. cotti* entered the Hawaiian Islands with poeciliid fishes from the Americas, and spread by translocation of poeciliid hosts across the archipelago for mosquito control. We also considered the alternative hypothesis of multiple independent co-introductions with host fishes originating from Asia. We inferred conduits of introduction and spread of *C. cotti* across the archipelago from geographic patterns of mtDNA sequence variation and allelic variation across 11 newly developed microsatellite markers. The distribution of haplotypes suggests that *C. cotti* spread across the archipelago following an initial introduction on O'ahu. Approximate Bayesian Computation modeling and allelic variation also indicate that O'ahu is the most likely location of introduction, from which *C. cotti* dispersed to Maui followed by spread to the other islands in the archipelago. Evidence of significant genetic structure across islands indicates that contemporary dispersal is limited. Our findings parallel historical records of non-native poeciliid introductions and suggest that remediating invasion hotspots could reduce the risk of infection in native stream fishes, which illustrates how inferences on parasite co-introductions can improve conservation efforts by guiding responses to emerging infectious disease in species of concern.

Introduction

Human-mediated movement of non-native parasites has resulted in a growing legacy of infectious diseases in wildlife (Daszak et al. 2000, Peeler et al. 2010, Poulin et al. 2011). The well-being of native host species can be improved by understanding pathways of colonization and spread of non-native parasites (Miura et al. 2006, Blum et al. 2007, Darling and Blum 2007, Biek and Real 2010). Epidemiological



monitoring, for instance, can be informed by inferring conduits of introduction and vectors of spread by identifying points of origin and movement potential. This in turn can guide implementation of management strategies, such as remediation of invasion hotspots or targeted control of key vectors to reduce infection risk, as well as preventative strategies such as prohibiting transport of vector species.

It is often stipulated that the origin and distribution of parasites depends on host movement (Criscione 2008), with the most mobile host exerting the greatest influence on parasites with complex life cycles (Nadler 1995). Discrepancies can arise, however, due to a number of factors that are not necessarily related to host dispersal that may influence infection and transmission, such as life history variation and the co-occurrence of cryptic species (Miura et al. 2006, Criscione et al. 2011, Gorton et al. 2012). Thus, inferring the origin and spread of non-native parasites requires more than assessing the distribution and movement of hosts (Miura et al. 2006, Biek and Real 2010, Lee et al. 2012).

Evaluating geographic patterns of genetic variation has proven to be a powerful approach for inferring the nature of species invasions, including the introduction location and spread of non-native parasites (Darling and Blum 2007, Le Roux and Wiczorek 2009). Estimates of genetic variation can help identify invasion routes, post-introduction admixture, and determine whether invasions have proceeded following multiple introductions (Blum et al. 2007, Rius et al. 2012, Stapley et al. 2015). Studies of genetic variation in non-native parasites have so far revealed complex colonization pathways, cryptic species, and founder effects (Miura et al. 2006, Blakeslee and Fowler 2012, Rey et al. 2015). For example, genetic analysis of an introduced trematode infecting the Asian mud snail *Batillaria attramentaria* revealed the presence of two cryptic parasite species with different conduits of colonization: one naturally dispersed by birds and the other by anthropogenic transport of intermediate snail hosts (Miura et al. 2006). Genetic analysis also can help identify characteristics that are associated with a parasite becoming invasive (Dudaniec et al. 2008, Gross et al. 2014). Parasites with high levels of gene flow, for instance, appear to have greater capacity to switch hosts (Greischar and Koskella 2007), which can contribute to establishment and success in novel environments (Garant et al. 2007, Greischar and Koskella 2007).

The co-introduction of parasites with non-native stream fishes is of particular concern (Vitule et al. 2009), especially on oceanic islands (Font 2003) where aquatic invasive species are a primary threat to endemic native stream fauna (Brasher 2003). Non-native species can dominate oceanic island stream assemblages, which otherwise feature low species diversity, high endemism, and strong representation of diadromous species sustained by ocean-stream connectivity (McDowall 2004, Ishikawa and Tachihara 2014). In streams on O'ahu (Hawai'i, USA), for example, the abundance and diversity of non-native fishes can be an order of magnitude higher than that of native fishes (Yamamoto and Tagawa 2000, Blum et al. 2014). Similarly, non-native parasites far

exceed all known parasites that have naturally colonized the archipelago (Font 2003), and native Hawaiian stream fishes are now predominantly infected by non-native parasites (Font 2003, Gagne et al. 2016).

The intestinal nematode *Camallanus cotti* has become the most widespread and abundant parasite of stream fishes across the Hawaiian Islands (Font 2003). The nematode is a generalist parasite with an indirect life cycle. Upon consumption, free-stage larvae infect cyclopoid copepods as their intermediate host. The parasite is transmitted to a range of definitive host fishes that ingest infected copepods. The life cycle is completed when mature female parasites within fish hosts release first stage larvae into surrounding waters (Font and Tate 1994, Font 2003).

Four of the five endemic Hawaiian stream fishes (*Awaous stamineus*, *Lentipes concolor*, *Eleotris sandwicensis*, and *Stenogobius hawaiiensis*, but not *Sicyopterus stimpsoni*) have become infected by *C. cotti* (Font 2003, Lindstrom et al. 2012). The pathology of *C. cotti* in native Hawaiian stream fishes is not known, but *C. cotti* is considered a threat because it causes disease (e.g. hemorrhage and erosion of the rectum mucosa) as well as behavioral changes (e.g. apathy and reduced libido) in other species (Kim et al. 2002, Font 2003, Menezes et al. 2006). Infection of native hosts by *C. cotti* increases with host size, declining precipitation, and declining water quality (Gagne et al. 2015, Gagne and Blum 2016). Recent archipelago-wide surveys of watersheds on all islands with perennial streams also indicate that infection of native hosts has become decoupled from the distribution of non-native hosts (Gagne et al. 2015). Some of the highest infection levels in native fishes are in remote watersheds that do not harbor introduced fishes, suggesting that *C. cotti* is capable of spreading independently of its original host (Gagne et al. 2015).

Introduced poeciliid fishes (guppies and allied genera), which are native to the Americas and Africa, are thought to have served as the primary vectors of *C. cotti* throughout the archipelago. It has been hypothesized that *C. cotti* arrived with poeciliids that were imported from the Americas for mosquito control (Yamamoto and Tagawa 2000, Font 2003). Though *C. cotti* is native to Asia (Font 2007), poeciliids could have been infected with *C. cotti* when introduced into Hawai'i, as infections are frequent in aquaculture stocks (Yamamoto and Tagawa 2000, Kim et al. 2002, Font 2003, 2007). Two introductions of poeciliids occurred on O'ahu in the early 20th century from the mainland United States: mosquito fish *Gambusia affinis* and sailfin molly *Poecilia latipinna* were introduced in 1905, and guppies *P. reticulata* and green swordtail *Xiphophorus helleri* were introduced in 1922 (Yamamoto and Tagawa 2000). Secondary introductions of poeciliids may have then spread *C. cotti* across the archipelago, beginning with Maui (Yamamoto and Tagawa 2000). Poeciliids can tolerate brackish conditions, which could allow for movement between some watersheds through estuarine environments, but movement between islands is almost certainly restricted to human mediated translocation (Yamamoto and Tagawa 2000).

There are several plausible invasion scenarios for *C. cotti* spreading across the archipelago. First, the prevailing hypothesis is that *C. cotti* was introduced to the Hawaiian Islands with poeciliid fishes from the Americas. Second, *C. cotti* could have been introduced to Hawai'i with non-native hosts from Asia. Some of the earliest introductions of non-native freshwater fishes to Hawai'i originated from Asia, most notably carp *Cyprinus carpio* and dojo *Clarias fuscus*, which were independently established in ponds on each island during the 19th century (Yamamoto and Tagawa 2000). Though less likely (Gagne et al. 2015), it is also possible that *C. cotti* was co-introduced with non-native fish (i.e. poeciliids or species originating from Asia) after which *C. cotti* quickly spread throughout the archipelago by secondarily infected, highly dispersive native hosts. Native Hawaiian stream fishes are euryhaline (i.e. species capable of surviving a wide range of salinities) or exhibit either obligate or facultative amphidromy (Hogan et al. 2014). Obligate amphidromous species mature and spawn in freshwater streams, but disperse through the ocean as larvae for up to six months. Facultative amphidromous species may bypass the marine phase in favor of remaining in freshwater throughout the life cycle (Hogan et al. 2014). Larvae of native amphidromous fishes are not infected with *C. cotti*, hence oceanic dispersal of native fish is not a likely conduit for transport of *C. cotti* between watersheds or islands (Font 2007). It is possible, however, that native euryhaline fish transport *C. cotti* between watersheds and islands (Gagne et al. 2015).

In this study, we examined genetic variation in *C. cotti* infecting an at-risk native Hawaiian stream fish to evaluate the prevailing narrative that the parasite was co-introduced with non-native poeciliid hosts (Yamamoto and Tagawa 2000). We tested the hypothesis that *C. cotti* was originally introduced to O'ahu and subsequently spread to other islands. If co-introduced and spread with poeciliids, *C. cotti* would be expected to exhibit signatures of a parallel invasion history corresponding to a single introduction and step-wise advance to Maui and then to other islands. If *C. cotti* was repeatedly co-introduced with Asian fish hosts, then it would instead exhibit patterns of genetic variation reflecting geographic differentiation among islands arising from multiple independent, concurrent founding events. If secondarily spread by highly dispersive native hosts, *C. cotti* would be expected to exhibit little if any signature of geographic differentiation.

We combined traditional population genetic analyses with Approximate Bayesian Computation to test competing hypotheses of single versus multiple introductions and spread of *C. cotti* across the archipelago. This enabled us to infer the most likely number, location, and progression of colonization events across the archipelago. By resolving the uncertain origins and dispersal history of *C. cotti*, we aimed to improve understanding of how non-native parasites with complex life cycles can succeed in novel environments. We also aimed to inform efforts to conserve imperiled endemic Hawaiian stream fishes, and in turn, provide a framework for assessing emerging infectious disease in other at-risk species of concern.

Methods

Sample collection and DNA extraction

In this study, we focused on *C. cotti* parasitizing the facultative amphidromous goby *Awaous stamineus*. Despite being the most widespread and abundant native stream fish in the archipelago, the species is of conservation concern and thus is afforded protection by the State of Hawai'i (Brasher 2003, Walter et al. 2012, Blum et al. 2014). Between 2009 and 2011, we sampled *A. stamineus* from 35 watersheds on the five Hawaiian Islands with perennial streams (Supplementary material Appendix 3 Table A1; Hawai'i, Maui, Moloka'i, O'ahu and Kaua'i, Fig. 1). A total of 935 *A. stamineus* (2009, $n = 557$; 2011, $n = 378$) were collected with hand nets from each watershed (Kaua'i, $n = 190$; O'ahu, $n = 265$; Moloka'i, $n = 192$; Maui, $n = 156$; Hawai'i, $n = 132$). All fish were necropsied for intestinal macroparasites with a dissecting microscope following Hoffman (1999). Parasites were preserved in 95% ethanol for DNA extraction. For collections made in 2009, a total of 278 fish were found to be infected; DNA was extracted from a single parasite from each infected fish. There were 162 infected fish in the 2011 collections; DNA was extracted from a total of 566 parasites, with ≤ 15 parasites examined per fish. No other endoparasite exhibited a prevalence greater than 2% (Gagne et al. 2015, 2016). DNA was extracted from whole male parasites and from the heads of female parasites using the Qiagen DNeasy Blood and Tissue extraction kit (Qiagen, Valencia, CA).

Mitochondrial DNA sequencing and analysis

A 427 base pair (bp) fragment of the mitochondrial cytochrome *c* oxidase I gene (*cox1*) was PCR-amplified from 241 *C. cotti* (one parasite per host) collected in 2009 using primers FCOX1A and RCOX1A (Wu et al. 2009; Supplementary material Appendix 1). Sequences were assembled and edited using Sequencher ver. 4.9 software (Gene Codes, Ann Arbor,

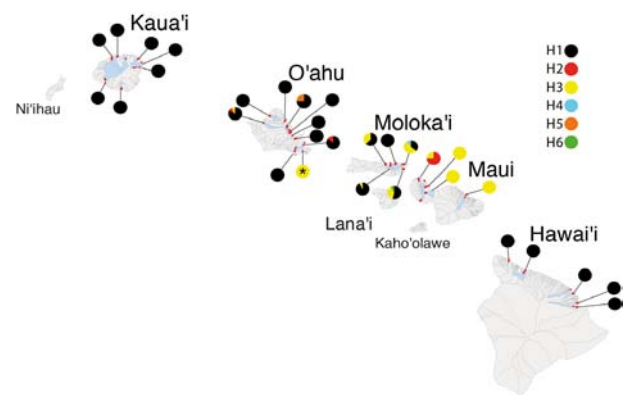


Figure 1. Distribution and relative frequency of *cox1* haplotypes of *Camallanus cotti* across the Hawaiian archipelago. Locations marked with an asterisk indicate watersheds with only one sample present. Red circles mark sampling sites.

MI). DNAsp ver. 5.10.1 (Librado and Rozas 2009) and Arlequin ver. 4.613 (Excoffier and Lischer 2010) were used to estimate the total number of mitochondrial haplotypes, haplotype diversity, pairwise differences among haplotypes, nucleotide diversity among haplotypes, and effective number of haplotypes for each island. Effective number of haplotypes was calculated as the reciprocal of the sum of squared frequencies. A neighbor joining tree with 1000 bootstrap replicates was constructed in MEGA ver. 6.0 (Tamura et al. 2013) to assess the presence of cryptic species and to infer relationships among haplotypes from the Hawaiian archipelago and the native range in China (Wu et al. 2009). Relationships among haplotypes also were reconstructed with a statistical parsimony network using Network ver. 2.1 (Polzin and Daneschmand 2003).

Analyses of molecular variance (AMOVAs) were carried out for estimating hierarchical genetic differentiation (Φ_{ST} , Φ_{SC} , Φ_{CT}) and the proportion of variance attributable to different hierarchical scales (island, watershed, within watershed). Pairwise Φ_{ST} values between watersheds and islands were also calculated. To account for multiple comparisons, p-values were corrected using the sequential Bonferroni method (Holm 1979). Both analyses were performed in Arlequin ver. 4.613 (Excoffier and Lischer 2010) and statistical significance was assessed according to 10 000 permutations.

Microsatellite development and analysis

A total of 844 individuals (2009, $n = 278$; 2011, $n = 566$) were genotyped at 14 microsatellite markers (Supplementary material Appendix 1) that were newly developed for *C. cotti* through paired-end Illumina sequencing following (Castoe et al. 2012). Electropherograms were scored and binned with

GeneMarker ver. 9.0 software (Softgenetics, State College, PA). To evaluate potential genotyping errors, all individuals with private alleles and a random subset of 75 individuals were genotyped twice. Data from three loci with only 1–2 alleles were discarded; data from the remaining 11 loci were used in all subsequent analyses (Table 1). To ensure that differences in sampling (i.e. among individuals, among sites) did not introduce analytical biases, all results were corroborated using a dataset with a single randomly selected parasite per fish.

Genepop ver. 4.2 (Rousset 2008) was used to test for Hardy–Weinberg Equilibrium and linkage disequilibrium. The number of alleles, observed and expected heterozygosity, and Shannon's information index were calculated for each island and watershed using GenAlex ver. 6.501 (Peakall and Smouse 2012). Arlequin ver. 4.613 (Excoffier and Lischer 2010) was used to calculate rarefied allelic richness and pairwise F_{ST} values, as well as to conduct AMOVAs. Pairwise comparisons were conducted between islands as well as between watersheds. Pairwise p-values between islands and between watersheds were corrected using the sequential Bonferroni method. Bayesian clustering software Structure ver. 2.1 (Pritchard et al. 2000) also was used to assess genetic structure with the admixture model, correlated allele frequencies (Pritchard et al. 2000, Hubisz et al. 2009) and allowing the parameter α to vary among populations. We ran 10 replicate analyses of the full data set with a burn-in period of 1 000 000 iterations followed by an additional 1 000 000 iterations and consecutive runs with K set from 1 to 10. Using Structure Harvester (Earl and vonHoldt 2012), the uppermost level of genetic structure was selected based on the rate of change of the log probability of data $[\ln P(D)]$ between successive K values (ΔK) following Evanno et al. (2005). To detect potential genetic

Table1. Microsatellite loci developed for *Camallanus cotti* used to assess genetic variation of parasites infecting *Awaous stamineus* in Hawai'i.

Locus	Primer sequences (5'–3')	Repeat motif	Dye label	Na	Allele size range (bp)	Ho (He)
CC-01	F_TTCTAAGATGATAGAGGTGCGAGC R_TGTACGTGAAGGTGATAACTGCC	AAAT (44)	HEX	9	192–338	0.463 (0.410)
CC-03	F_AAGAACTGACCGGAAATCGC R_AACTGGAATGAGTTGAGAGCCC	AATG (36)	FAM	8	189–217	0.993 (0.526)
CC-04	F_TTCTCGTATAAATAAGTCCAGTGCC R_AAATGATAAGAAACCCTCATTCC	AGTG (36)	FAM	6	472–504	0.248 (0.289)
CC-06	F_CATAAAGCATTGTCTGCCTATTGC R_CCGATCAAGTGAATATTATGAAATGG	AATG (32)	TAMRA	6	164–232	0.208 (0.183)
CC-08	F_AATCGAGCGAGAAGAGAGGG R_TCCTCAATGTATGAAACTCATCC	AAAG (28)	HEX	6	220–240	0.110 (0.161)
CC-09	F_TTCGTGAACGCAACCTTCC R_AGAAGACGAATAGCATATATTGTGTGG	ATGG (28)	FAM	6	276–296	0.225 (0.210)
CC-10	F_TCATCGGAATCTAATATCGAATGC R_TGGAAAGGTTCTGCTTCTTATCG	ATAC (28)	TAMRA	4	172–192	0.513 (0.469)
CC-12	F_AATCGAGCGAGAAGAGAGGG R_CACTTGCATTAAATTAGTCTACATGGG	AAAG (28)	HEX	6	142–160	0.022 (0.041)
CC-14	F_TTTATATAGCAGATATGTTTCTGTTGC R_CCCAAAGAAATGATGAAATCTGC	AAAT (32)	HEX	6	274–290	0.224 (0.271)
CC-17	F_GAAGGCGAGAGCATAAAGGC R_ATTCATTGTTGCCTTGCGG	AAAT (36)	FAM	9	142–232	0.205 (0.250)
CC-20	F_TGGTATGCCACAAAGTTCGC R_CATGAAACATTACAATTTACAATGGC	AAAT (36)	TAMRA	3	179–191	0.487 (0.423)

structure within the inferred genetic demes identified from the full data set, subsets of data representing each genetic deme were independently analyzed as described above. Once K values were estimated, the final Q coefficients for each population were obtained by averaging the 10 runs of the selected K in CLUMPAK using the default parameters including the 'large K greedy' algorithm and 2000 random input order repeats (Kopelman et al. 2015). Additionally, discriminant analysis of principal components (DAPC) was used to evaluate the extent of genetic structure among the sampled locations (Jombart et al. 2010). DAPC was run in R 3.1.1 (R Core Team) using the 'adegenet' package (Jombart 2008).

Testing models of invasion history

Competing colonization scenarios were evaluated using approximate Bayesian computation (ABC) with the program

DIYABC ver. 2.0 (Cornuet et al. 2014). Models were first run using microsatellite data and subsequently run using a combined microsatellite and mitochondrial dataset. We evaluated six alternative scenarios of colonization that reflect potential routes of colonization based upon introduction records of non-native fishes to Hawai'i and information from *cox1* haplotype distributions (Fig. 2). The scenarios tested were: (scenario 1) initial colonization of O'ahu followed by sequential colonization of all other islands in order of geographic proximity to O'ahu; (scenario 2) two independent colonization events, one on O'ahu from which Kaua'i and Hawai'i were colonized, and a second event on Maui from which Moloka'i was subsequently colonized; (scenario 3) simultaneous and independent colonization of all islands; (scenario 4) initial colonization of O'ahu followed by Maui, and where Moloka'i was then colonized from Maui, and where O'ahu was the source of parasites on Kaua'i and Hawai'i; (scenario 5) initial colonization of Maui followed

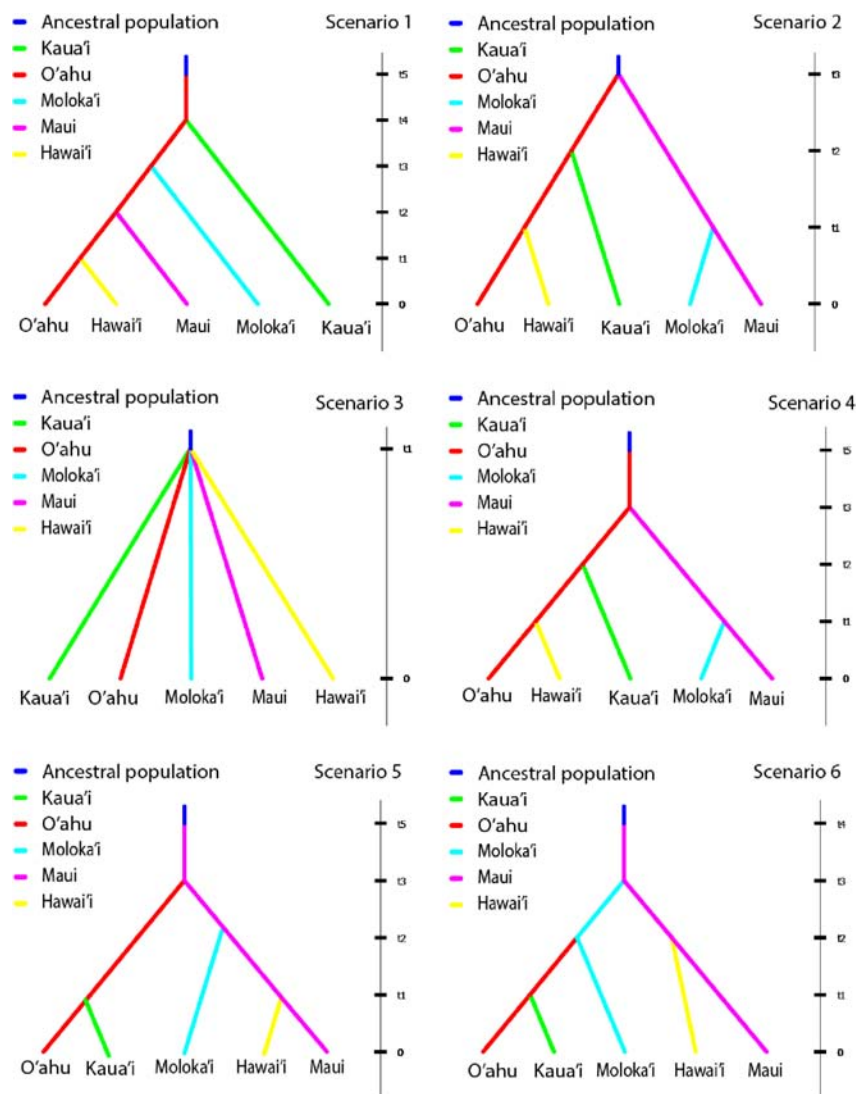


Figure 2. Competing colonization scenarios of *Camallanus coti* into Hawai'i that were evaluated using approximate Bayesian computation.

by O'ahu, and where Moloka'i and Hawai'i were colonized from Maui and where O'ahu was the source of parasites on Kaua'i; (scenario 6) initial colonization of Maui followed by stepwise colonization of other islands (i.e. Maui to Moloka'i and Hawai'i, Moloka'i to O'ahu, and O'ahu to Kaua'i). Simulations for each scenario were run twice, first setting a constraint for colonization times to the last 200 yr (i.e. the maximum time since poeciliid introductions), and a second set of runs considering colonization events over the last 1000 yr (i.e. to account for potential introductions by Polynesian colonists). Posterior probabilities of the alternative invasion scenarios were calculated using 1% simulated datasets to estimate the relative posterior probability of each scenario with a logistic regression over a linear discriminant analysis of the summary statistics (Estoup et al. 2012). Posterior predictive error of the alternative invasion scenarios (i.e. the proportion of wrongly identified scenarios) was calculated using 500 simulated datasets closest to the observed dataset for the direct approach (i.e. the number of times the right scenario has the highest posterior probability), as well as using 1000 simulated datasets for the logistic approach, following the same procedure for estimating their respective posterior probabilities (Estoup et al. 2012, Cornuet et al. 2014).

Data deposition

Data available from the Dryad Digital Repository: <<http://dx.doi.org/10.5061/dryad.b9h54>> (Gagne et al. 2017).

Results

Mitochondrial DNA variation

We found low levels of haplotype diversity and an uneven distribution of *cox1* haplotypes across the Hawaiian

archipelago. Only six haplotypes were found among the 241 individuals sampled across all five islands, as compared to the recovery of 74 haplotypes from 152 *C. cotti* sampled in the native range (Wu et al. 2009; Fig. 3). None of the haplotypes recovered in Hawai'i were shared with those sampled in the native range. The NJ tree recovered all native haplotypes as a paraphyletic group with respect to the Hawaiian haplotypes, which formed a monophyletic clade with moderate statistical support (Supplementary material Appendix 3 Fig. A1). The most common *cox1* haplotype (H1) in the Hawaiian archipelago was recovered in 75% of individuals and was present on all islands except Maui (Fig. 1 and 3, Supplementary material Appendix 3 Table A1). O'ahu and Moloka'i were found to harbor the greatest number of haplotypes with four haplotypes present on each island (Supplementary material Appendix 3 Table A1). Moloka'i harbored *C. cotti* with the highest haplotype and nucleotide diversity (Table 2), as well as the highest average pairwise differences among haplotypes (Table 2).

We recovered evidence of genetic differentiation among islands but not among the majority of watersheds (Table 3). Global genetic variance among islands was $\Phi_{CT} = 0.507$ ($p < 0.0001$). Pairwise Φ_{ST} values between islands ranged from 0.00 to 0.952. Populations on Maui were significantly different from populations on all other islands (Table 4). Populations on Maui also exhibited the highest pairwise Φ_{ST} values ($\Phi_{ST} \geq 0.527$) followed by populations on Moloka'i, which were significantly different from populations on all other islands except those on Hawai'i (Table 4). Pairwise Φ_{ST} values between watersheds ranged from 0.00 to 1.00. Most pairwise estimates of genetic differentiation among watersheds were insignificant after sequential Bonferroni correction, but significant comparisons were found between Maui and Moloka'i watersheds and watersheds on all other islands.

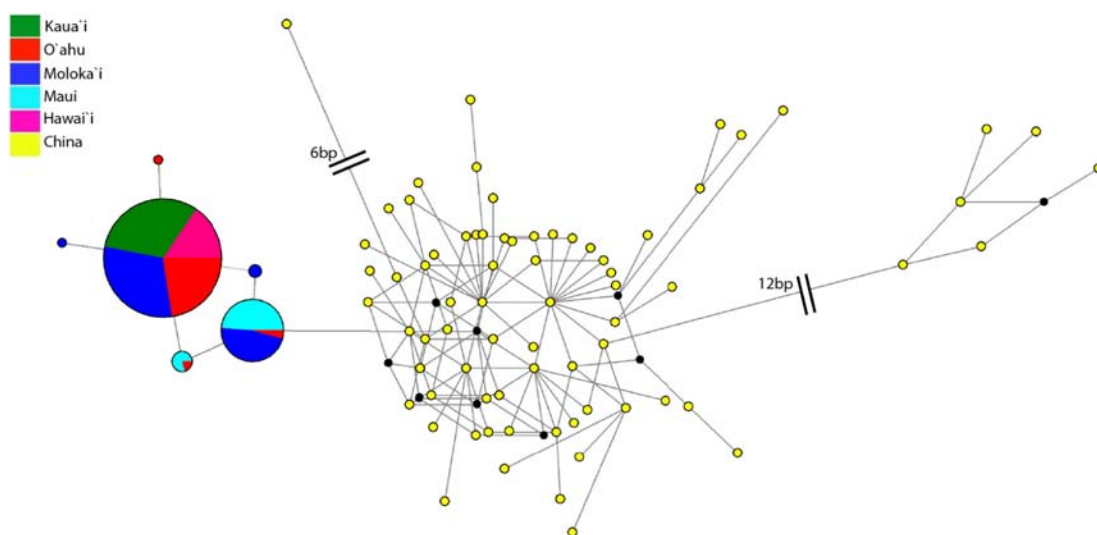


Figure 3. Haplotype network from *cox1* sequences of *Camallanus cotti* across the Hawaiian archipelago as well as the native range of China from the Yangtze, Pearl and Mingjiang Rivers (sequences from China are taken from Wu et al. 2009). Circle size reflects the relative abundance of haplotypes recovered among the locations sampled in the Hawaiian Islands. Haplotypes from China are not sized to reflect abundance.

Table 2. Island level diversity measures of *Camallanus cotti*. n = sample size, h = the number of mitochondrial haplotypes, Hd = haplotype diversity, Pairwise D = average pairwise differences among haplotypes, π = nucleotide diversity among haplotypes, k = effective number of haplotypes, Na = average number of alleles, Ar = rarefied allelic richness, I = Shannon's information index, He = expected heterozygosity, and Ho = observed heterozygosity.

mtDNA <i>cox1</i>	n	h	Hd	Pairwise D	π	k
Kaua'i	54	1	0	0	0	1
O'ahu	43	4	0.174	0.271	0.002	1.206
Moloka'i	80	4	0.463	0.889	0.008	1.844
Maui	39	2	0.254	0.259	0.002	1.324
Hawai'i	25	1	0	0	0	1
nDNA microsatellites	n	Na	Ar	I	He	Ho
Kaua'i	178	4.091	3.30	0.432	0.225	0.210
O'ahu	131	4.000	3.57	0.594	0.333	0.263
Moloka'i	268	4.636	2.85	0.429	0.244	0.331
Maui	127	3.545	3.06	0.555	0.332	0.438
Hawai'i	139	3.727	3.22	0.605	0.358	0.440

Microsatellite allelic variation

A total of 783 individuals (2009, $n = 233$; 2011, $n = 550$) were included in all analyses following removal of individuals with missing data from more than four loci. The 11 loci scored for all sampled individuals exhibited an average of 6.3 alleles per locus and a range of 3 to 9 alleles per locus. Average Shannon index values were 0.508 ± 0.037 with a range of 0.119 to 0.992, and observed heterozygosity ranged from 0.022 to 0.993 across loci (Table 2). Across all islands, observed heterozygosity (0.331 ± 0.083) was nearly equal to expected heterozygosity (0.333 ± 0.054), but observed heterozygosity was greater than expected heterozygosity on Moloka'i, Maui, and Hawai'i but less than expected heterozygosity on O'ahu and Kaua'i (Table 2). Evidence of linkage disequilibrium was found in only 19 of 275 comparisons (6.9%); comparisons between the same pairs of loci were not consistently significant across islands. The results from analyses run with one parasite per sample paralleled those of the full data set. Accordingly, only the results from the full data set are presented.

Populations on Moloka'i exhibited the greatest number of alleles but the lowest number of rarefied allelic richness and the lowest Shannon index values. This is, in part, a consequence of Moloka'i harboring the greatest number of private alleles. In contrast, populations on Hawai'i had the fewest private alleles of any island, even though these populations also exhibited the highest Shannon index values and the highest observed and expected heterozygosity values.

Populations on O'ahu also exhibited comparably high Shannon index values and heterozygosity (Table 2), as well as the highest level of rarefied allelic richness (Table 2).

Analysis of multilocus nuclear genotypic variation recovered high levels of among-island differentiation (global $F_{ST} = 0.221$, $p < 0.001$) consistent with patterns of mtDNA variation. Pairwise F_{ST} values between watersheds ranged from 0.009 to 0.373, although the majority of comparisons were not statistically significant following sequential Bonferroni correction. Nonetheless, corrected pairwise F_{ST} values were significantly different between all islands ($p < 0.01$) with the greatest F_{ST} value occurring between Moloka'i and Kaua'i ($F_{ST} = 0.264$) and the lowest between Moloka'i and Maui ($F_{ST} = 0.041$; Table 3). Genetic variation was significant at all hierarchical levels ($p < 0.001$; Table 3).

Bayesian cluster analysis of multi-locus genotypes did not recover a clear peak or reach an asymptote in posterior probabilities [$\ln P(D)$], however an increase in variation and fluctuating values were observed among runs beyond $K = 6$ (Supplementary material Appendix 3 Fig. A2). The highest rate of change of the posterior probabilities (ΔK) clearly indicated the uppermost level of genetic structure occurred at $K = 2$. Runs at $K = 2$ recovered one cluster (K1) consisting of individuals from Maui and Moloka'i with high proportions of membership (Q values > 0.900), and another cluster (K2) consisting of individuals from Hawai'i, Kaua'i (Q values > 0.980), and from O'ahu. Individuals from O'ahu exhibited greater admixture with the other cluster ($Q = 0.814$) (Fig. 4). The inferred structure was congruent

Table 3. Analysis of molecular variance (AMOVA) with reference to the percentage of variation across three hierarchical spatial scales based on *cox1* and microsatellite allelic variability in *Camallanus cotti*.

	df	Sum of squares	Variance components	Percentage of variation	p
mtDNA <i>cox1</i>					
Among islands	4	39.8	0.20604 Va	50.73	< 0.001
Among watersheds within islands	26	13.2	0.05046 Vb	12.42	< 0.001
Within watersheds	205	30.7	0.14962 Vc	36.84	< 0.001
nDNA microsatellites					
Among islands	4	289.9	0.20697 Va	16.90	< 0.001
Among watersheds within islands	30	113.1	0.06438 Vb	5.26	< 0.001
Within watersheds	1629	1553.3	0.95354 Vc	77.85	< 0.001

Table 4. Genetic differentiation of *Camallanus cotti* between islands, based on *cox1* (Φ_{ST} , above diagonal) and microsatellite allelic (F_{ST} , below diagonal) variation. Bold values represent significant values after sequential Bonferroni correction.

	Kaua'i	O'ahu	Moloka'i	Maui	Hawai'i
Kaua'i		0.040	0.246	0.953	0.000
O'ahu	0.125		0.137	0.853	0.014
Moloka'i	0.264	0.180		0.527	0.196
Maui	0.248	0.114	0.041		0.930
Hawai'i	0.167	0.102	0.243	0.196	

with the distribution of mitochondrial haplotypes; the frequency of mtDNA haplotypes was significantly related to the recovered genetic clusters ($\chi^2(5, n = 236) = 70.314$, $p < 0.00001$). When the two groups of islands conforming to the main clusters were analyzed independently, K1 showed a clear peak at $K = 2$ in both $\ln P(D)$ and ΔK values. However, $> 93\%$ of the individuals were assigned with a proportion of ancestry $\geq 85\%$ to the same cluster (Q values ≥ 0.946 ; Fig. 4). In cluster K2, on the other hand, the highest $\ln P(D)$ values were observed between $K = 7$ and $K = 8$, and ΔK showed the uppermost level of genetic structure at $K = 5$. Two of the inferred clusters exhibited low membership proportions in all islands (Q values ≤ 0.086). Each of the other three clusters was preferentially assigned to each of the islands in this group, but with differing levels of admixture. Higher proportions of membership were observed for individuals from Kaua'i ($Q = 0.665$) and Hawai'i ($Q = 0.641$) than O'ahu ($Q = 0.481$), which exhibited a signature of admixture with Kaua'i (Fig. 4).

Discriminant analysis using DAPC identified three overlapping genotypic clusters (Fig. 5). A central cluster consisted of partially overlapped individuals from Kaua'i and O'ahu. A second cluster consisted of individuals from Moloka'i and Maui, with individuals from Moloka'i slightly overlapping more with the central cluster composed of individuals from Kaua'i and O'ahu. The most differentiated and distinct cluster consisted of individuals from Hawai'i (Fig. 5).

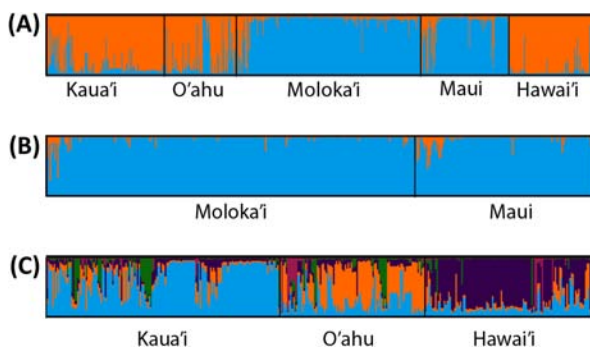


Figure 4. (A) Bayesian estimate of population structure at $K = 2$ for all islands based on microsatellite variation from *Camallanus cotti* across the Hawaiian archipelago. (B and C) Bayesian estimate of population structure within clusters identified during analysis with all samples at $K = 2$ and $K = 5$, respectively.

Testing models of invasion history

The greatest support was recovered for scenario 4 in all analyses (i.e. based on the microsatellite data and the combined dataset, constraining the colonization time to 200 and 1000 yr; Fig. 2). Scenario 4 represents an initial colonization of O'ahu and early colonization of Maui, where Maui serves as a source for colonization of Moloka'i. Scenario 4 was the best model of colonization according to a logistic regression on the linear discriminant analysis of the summary statistics (colonization time 200 yr PP = 0.995, CI = 0.989–1.000; colonization time 1000 yr PP = 0.995, CI = 0.988–1.000). The confidence intervals also did not overlap with those of the other scenarios tested. For the models constraining colonization time to 200 yr, the overall posterior predictive error was 24.8% following the direct approach and 7.8% following the logistic approach. For the models constraining colonization time to 1000 yr, the overall posterior predictive error was 34.1% following the direct approach and 26.8% following the logistic approach. The least likely invasion history was scenario 3, in which all islands were independently colonized.

Discussion

Introductions and invasion pathways

Multilocus genetic analysis of the non-native nematode parasite *Camallanus cotti* supports the hypothesis of a single introduction event to O'ahu followed by secondary spread across the Hawaiian archipelago. This inference highlights how translocation of non-native species can promote the spread of pathogens in species of conservation concern. Our results indicate that *C. cotti* did not follow a linear stepwise invasion of the islands originating from one introduction site. Patterns of variation and our best-fit colonization scenario indicate that an initial introduction took place on O'ahu from which the parasite colonized Maui, after which both populations served as independent sources for introductions to the other islands; O'ahu serving as the primary source to Kaua'i and Hawai'i, and Maui serving as the primary source to Moloka'i. This suggests that widespread infection of native stream fishes was not a consequence of transmission from independent introductions of Asian fishes throughout the archipelago (Font 2003) or a consequence of transmission by highly dispersive native fishes (Gagne et al. 2015). Rather, it is consistent with the prevailing narrative of co-introduction and spread with infected poeciliid hosts, which were initially introduced to O'ahu from the mainland United States and then translocated to Maui for mosquito control (Yamamoto and Tagawa 2000).

Patterns of genetic diversity also appear to reflect initial colonization and pathways of spread through co-introduction with non-native hosts (Gandon and Michalakis 2002, Greischar and Koskella 2007). Evidence of a single monophyletic clade of *C. cotti* in Hawai'i relative to the native range is

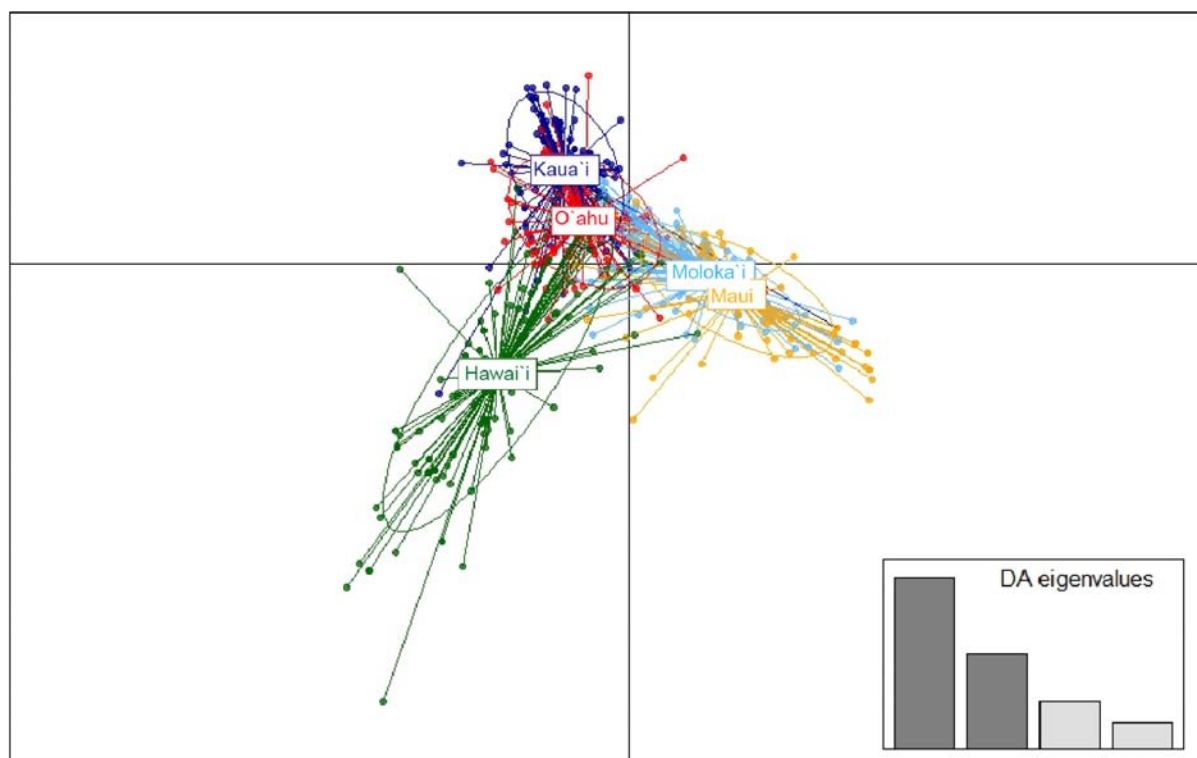


Figure 5. Discriminant analysis of principal components scatterplot of the first two principal components describing the genetic clusters of *Camallanus cotti* based on microsatellite allelic variation. Each point represents an individual colored-coded according to island of origin (Kaua'i = blue, O'ahu = red, Moloka'i = light blue, Maui = orange, Hawai'i = green). Insert graph represents the eigenvalues of the first four principal components, with the dark grey bars identifying the first two discriminant functions used in the scatterplot.

more consistent with a single introduction than multiple independent introductions. Relatively low levels of genetic diversity (Picard et al. 2004, Wu et al. 2009) across the archipelago likely reflect conditions during or following colonization, such as the establishment of a small founding population or a population bottleneck following a founding event (Reed and Frankham 2003). Populations on O'ahu and Maui exhibited comparably higher levels of mitochondrial and nuclear genetic diversity relative to those on other islands, consistent with the expectation that the original location of an introduction and sites of early colonization often harbor the highest levels of genetic diversity while secondary invasion sites harbor lower levels due to successive founder events (Rey et al. 2015). The relatively high diversity and excess of heterozygotes on Moloka'i, on the other hand, possibly reflects contributions and admixture from Maui and O'ahu, which harbor different genetic clusters and haplogroups. It is also possible that higher levels of diversity occur on Moloka'i because it harbors populations of *A. stamineus* infected with the greatest numbers of parasites (Gagne et al. 2015). Unexpectedly high levels of diversity detected on Hawai'i, particularly compared to Kaua'i, may be a signature of the sequence of colonization ending with *C. cotti* arriving on Hawai'i. Populations on an invasion front can exhibit greater genetic diversity due to genetic drift coupled with smaller population sizes resulting in the increased frequency of rare alleles, or the establishment

of novel alleles that arise with new mutations (Klopfstein et al. 2006, Gross et al. 2014).

Evidence that the observed patterns in *C. cotti* depart from those observed in the most vagile hosts in Hawaiian streams further suggests that spread has resulted from co-introduction with non-native hosts. Population connectivity in parasites often matches that of the most vagile host species (Nadler 1995). Yet, unlike *C. cotti*, endemic amphidromous fishes exhibit little to no genetic structure (McDowall 2003, Moody et al. 2015, Alda et al. 2016). This incongruence has likely arisen because larvae of native amphidromous fishes, which are capable of moving among watersheds and islands via marine dispersal, are not infected by *C. cotti*. The presence of *C. cotti* in remote watersheds that do not harbor any non-native fishes suggests, however, that the genetic structure of the parasite across Hawai'i may be influenced by native euryhaline fish hosts (e.g. *Kuhlia* flagtails) that routinely move between marine and freshwater environments (Gagne et al. 2015). Comparing the genetic structure of *C. cotti* with that of prevalent non-native hosts (Purcell and Stockwell 2015) and native euryhaline fish hosts would help clarify the extent to which parasite dispersal is contingent on transport through marine environments. Though improbable, it is also possible that phoretic transport of intermediate copepod hosts on aquatic birds or mammals influences the population structure of *C. cotti* (Frisch et al. 2007, Gagne et al. 2015).

Conservation and management implications

Determining the genetic structure of an introduced parasite can help define the optimal geographic scale for control or mitigation. Here, Bayesian clustering, DAPC, and AMOVA results indicate that the genetic structure of *C. cotti* across Hawai'i is largely attributable to differentiation among islands, reflecting a sequence of initial arrival, subsequent spread across watersheds within each island, and restricted movement among islands. Some patterns of regional differentiation nonetheless correspond to geographical proximity of islands. For example, Bayesian clustering indicated higher gene flow or more recent connectivity between Moloka'i and Maui than among other neighboring islands. Signatures of admixture, particularly in populations on O'ahu, provide further evidence of connectivity among neighboring islands. Yet we also found evidence of greater connectivity among geographically disjunct islands. For instance, Hawai'i grouped with Kaua'i and O'ahu rather than more geographically proximate islands. This supports the idea that spread among islands is attributable to human-mediated dispersal (Helmus et al. 2014), with O'ahu having served as a source of poeciliids to stock other islands for mosquito control because Honolulu is a transportation hub for the state of Hawai'i.

Evidence that *C. cotti* exhibits genetic structure at the island but not watershed level suggests that mitigation efforts could be successful if focused on containing or controlling conditions island by island. This finding also suggests that, barring efforts to control further spread, it is likely *C. cotti* will become established in watersheds where it is not currently present (Gagne et al. 2015). Impeding further translocation of non-native hosts, as opposed to reducing densities of non-native hosts through direct removal (Gagne et al. 2016), or re-establishing hydrological conditions that favor native fishes (i.e. environmental flow regimes like pulsed high flows comparable to freshets) in watersheds with regulated surface flow could constitute key steps toward reducing non-native parasite burdens in native species of concern across Hawai'i (Gagne and Blum 2016).

Although it is difficult to reconstruct the exact sequence of parasite invasions, this study affirms the value of genetic approaches for inferring invasion pathways and potential for spread of non-native parasites infecting species of concern. Our findings illustrate that molecular epidemiological frameworks developed for other pathogens (Biek and Real 2010) can be useful for responding to emerging parasitic diseases, which have thus far been largely overlooked (Thompson et al. 2010). As has been shown for viral pathogens, for example, genetic analysis of introduced parasites can provide valuable information about the origins and emergence of parasitic disease, particularly when historical records are scarce or unclear (Rey et al. 2015). Nonetheless, inferential and statistical analyses of genetic variation have limitations. For example, DIYABC only identifies the best scenario from a selection

of plausible scenarios (i.e. as opposed to all possible scenarios). Assessing other more complex scenarios – including the possibility that *C. cotti* were introduced with Asian fishes to O'ahu and then subsequently dispersed via poeciliid hosts across the archipelago – will require further study such as comparative analyses of genetic variation in *C. cotti* from elsewhere in the world. Exploratory application and evaluation of other complementary approaches – including landscape and genomic methods – could also further expand capacity for responding to emerging parasitic diseases.

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Supplementary material (Appendix ECOG-02855 at <www.ecography.org/appendix/ecog-02855>). Appendix 1–3.